

## MPP-620AL-XF

A highly engineered polyethylene/aluminum oxide nanocomposite that provides slip, abrasion/rub resistance, and antiblocking

### Features and Benefits

- Maximum scratch and scuff resistance with slip and lubricity
- HDPE composite reinforced with 300 nm aluminum oxide nanoparticles (Mohs Hardness 9)
- Easier-to-disperse nanotechnology in a safe, non-nano powder
- Composite particle density maximizes efficiency and minimizes settling
- Effective replacement for PTFE additives

### Composition

High density polyethylene/aluminum oxide

### Recommended Addition Levels

0.5-1.5% (on total formula weight)

### Systems and Applications

Water based, solvent based and energy curable coatings and inks. Industrial coatings (including plastic and metal); stains, sealers and varnishes; wood coatings; printing inks and OPV's (including flexo and gravure); powder coatings; coil coatings; rubber additives.

### Typical Properties\*

|                            | <u>MPP-620AL-XF</u> |
|----------------------------|---------------------|
| Mean Particle Size (µm)    | 3.5 - 5.5           |
| Maximum Particle Size (µm) | 15.56               |
| Melting Point °C           | 120 - 124           |
| Density @ 25 °C (g/cc)     | 1.00                |
| NPIRI Grind                | 1.0 - 2.0           |

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## MPP-611AL

A highly engineered HDPE/aluminum oxide nanocomposite for unsurpassed scratch and scuff resistance

### Features and Benefits

- Maximum scratch and scuff resistance with slip and lubricity
- HDPE composite reinforced with 300 nm aluminum oxide nanoparticles (Mohs Hardness 9)
- Easier-to-disperse nanotechnology in a safe, non-nano powder
- Composite particle density maximizes efficiency and minimizes settling
- Effective replacement for PTFE additives

### Composition

HDPE/aluminum oxide

### Recommended Addition Levels

0.5-1.5% (on total formula weight)

### Systems and Applications

Water based, solvent based and energy curable coatings and inks. Industrial coatings (including plastic and metal); stains, sealers and varnishes; wood coatings; printing inks and OPV's (including flexo and gravure); powder coatings; coil coatings; rubber additives.

### Typical Properties\*

|                            | <u>MPP-611AL</u> |
|----------------------------|------------------|
| Mean Particle Size (µm)    | 3.5 - 5.5        |
| Maximum Particle Size (µm) | 15.56            |
| Melting Point °C           | 110 - 116        |
| Density @ 25 °C (g/cc)     | 0.99             |
| NPIRI Grind                | 1.0 - 2.0        |

This product is also available as a water based wax dispersion - Microspersion 611AL-50

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### GraphShield 730

Micronized composite of synthetic wax and edge functionalized graphene (EFG) for improved anticorrosion in powder coatings

#### Features and Benefits

- Improves corrosion resistance on bare and treated cold rolled steel (CRS) when formulated with titanium dioxide or other inorganic pigments
- Black wax composite delivers graphene in an easy to use micronized powder
- Formulated with Asbury Carbons EFG technology
- 1% typically reduces mm creep by 25%
- Must be added to the premix, ideally processed above 110 ° C

#### Composition

Graphene/synthetic wax

#### Recommended Addition Levels

1.0-3.0% (on total formula weight)

#### Systems and Applications

Powder coatings, industrial coatings.

### Typical Properties\*

|                            | <u>GraphShield 730</u> |
|----------------------------|------------------------|
| Mean Particle Size (µm)    | 8.0 - 12.0             |
| Maximum Particle Size (µm) | 31.00                  |
| Melting Point °C           | 108 - 113              |
| Density @ 25 °C (g/cc)     | 1.24                   |

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## MPP-620

High performance micronized polyethylene wax which provides a unique combination of surface slip, abrasion/rub resistance, and antiblocking in a wide variety of coatings and inks

### Features and Benefits

- Imparts excellent abrasion, rub and mar resistance with good surface slip
- Provides excellent antiblocking properties
- Adds heat resistance
- Better resistance to solvent absorption and swelling when compared to Synthetic wax

### Composition

High density polyethylene

### Recommended Addition Levels

1.0-3.0% (on total formula weight)

### Systems and Applications

Water based, solvent based and energy curable coatings and inks. Industrial coatings (including plastic and metal); stains, sealers and varnishes; wood coatings; printing inks and OPV's (including flexo and gravure); powder coatings; interior and exterior can and container coatings; coil coatings.

### Typical Properties\*

|                                   | <u>MPP-620VF</u> | <u>MPP-620XF</u> | <u>MPP-620XXF</u> |
|-----------------------------------|------------------|------------------|-------------------|
| <b>Mean Particle Size (µm)</b>    | 5.0 - 7.0        | 4.5 - 5.5        | 4.25 - 4.75       |
| <b>Maximum Particle Size (µm)</b> | 22.00            | 22.00            | 12.00             |
| <b>Melting Point °C</b>           | 120 - 124        | 120 - 124        | 120 - 124         |
| <b>Density @ 25 °C (g/cc)</b>     | 0.96             | 0.96             | 0.96              |
| <b>NPIRI Grind</b>                | 2.0 - 3.0        | 1.0 - 2.0        | 1.0 - 1.5         |

This product is also available as a water based wax dispersion - Microspersion 620VF-50

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## MP-22

High performance, versatile grade of micronized synthetic wax for improved slip, scratch and rub resistance in a wide variety of paints, coatings and inks

### Features and Benefits

- Imparts a high degree of lubricity and slip with rub and mar resistance
- Provides antiblocking properties
- Gives long-lasting water beading and weather resistance to exterior alkyd-based stains
- Consider MP-22C & MP-28C (spherical) for improved surface slip
- Biodegradable to OECD 302C test standard
- Not a microplastic per ECHA definitions

### Composition

Synthetic wax

### Recommended Addition Levels

1.0-3.0% (on total formula weight)

### Systems and Applications

Water based, solvent based and energy curable coatings and inks. Industrial coatings (including plastic, metal and masonry); stains, sealers and varnishes; wood coatings; printing inks and OPV's (including flexo and gravure); powder coatings; interior and exterior can and container coatings; coil coatings; powdered metals.

### Typical Properties\*

|                                   | <u>MP-22</u> | <u>MP-22VF</u> | <u>MP-22XF</u> | <u>MP-22XXF</u> |
|-----------------------------------|--------------|----------------|----------------|-----------------|
| <b>Mean Particle Size (µm)</b>    | 7.0 - 10.0   | 6.0 - 8.0      | 4.5 - 6.5      | 3.75 - 5.75     |
| <b>Maximum Particle Size (µm)</b> | 31.00        | 22.00          | 22.00          | 15.56           |
| <b>Melting Point °C</b>           | 110 - 115    | 110 - 115      | 110 - 115      | 110 - 115       |
| <b>Density @ 25 °C (g/cc)</b>     | 0.93         | 0.93           | 0.93           | 0.93            |
| <b>NPIRI Grind</b>                | 4.0 - 6.0    | 2.0 - 3.5      | 1.5 - 3.0      | 1.0 - 2.3       |

This product is also available as a water based wax dispersion - Microspersion 22-50

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### PolyGlide 1226XF

A highly engineered HDPE/ceramic/nanoceramic composite for abrasion resistance and lubricity

#### Features and Benefits

- HDPE composite reinforced with hard, inert ceramic microspheres and nanoceramic platelets
- Improved Taber abrasion resistance when compared to PE/PTFE additives
- Provides slip and lubricity
- Ideal for can and container coatings; 21CFR 175.300 approved
- Effective replacement for PTFE additives
- Compare to (coarser) Polyfluor 900

#### Composition

HDPE/ceramic

#### Recommended Addition Levels

0.5-1.5% (on total formula weight)

#### Systems and Applications

Water based, solvent based and energy curable coatings and inks. Industrial coatings (including plastic and metal); stains, sealers and varnishes; wood coatings; printing inks and OPV's (including flexo and gravure); powder coatings; can, container, and coil coatings; rubber additives.

#### Typical Properties\*

|                            | <u>PolyGlide 1226XF</u> |
|----------------------------|-------------------------|
| Mean Particle Size (µm)    | 3.5 - 5.5               |
| Maximum Particle Size (µm) | 15.56                   |
| Melting Point °C           | 109 - 115               |
| Density @ 25 °C (g/cc)     | 0.99                    |
| NPIRI Grind                | 1.0 - 2.0               |

This product is also available as a water based wax dispersion - Microspersion 1226XF-50

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## MicroKlear 418AL

Aluminum oxide modified carnauba wax for maximum scratch resistance with surface slip, clarity and gloss retention. Also useful as an additive in seed coatings

### Features and Benefits

- Carnauba wax composite reinforced with 300 nm aluminum oxide nanoparticles (Mohs Hardness 9)
- Provides maximum scratch resistance with surface lubricity
- Does not adversely affect clarity or gloss
- Easier-to-disperse nanotechnology in a safe, non-nano powder
- Composite wax/aluminum oxide particle is less abrasive on processing equipment compared to free aluminum oxide
- Effective replacement for PTFE additives

### Composition

Carnauba wax/aluminum oxide

### Renewable Carbon Index

>95%

### Recommended Addition Levels

1.0-3.0% (on total formula weight)

### Systems and Applications

Water based, solvent based and energy curable coatings and inks. Industrial coatings (including plastic, metal and leather); wood coatings; printing inks and OPV's (including flexo and gravure); interior and exterior can and container coatings; coil coatings; seed coatings.

### Typical Properties\*

|                            | <u>MicroKlear 418AL</u> |
|----------------------------|-------------------------|
| Mean Particle Size (µm)    | 6.0 - 8.0               |
| Maximum Particle Size (µm) | 22.00                   |
| Melting Point °C           | 81 - 86                 |
| Density @ 25 °C (g/cc)     | 1.04                    |
| NPIRI Grind                | 2.0 - 3.5               |

This product is also available as a water based wax dispersion - Microspersion 418AL-40

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## Superslip 6515AL

A highly engineered HDPE/EBS/aluminum oxide nanocomposite for maximum scratch, scuff and block resistance

### Features and Benefits

- Maximum scratch resistance; equal to PE/PTFE additives
- HDPE/EBS composite reinforced with 300 nm aluminum oxide nanoparticles (Mohs Hardness 9)
- Good abrasion resistance with slip and lubricity
- Amide wax component provides excellent antiblocking and release properties
- Ideal for can and container coatings; 21CFR 175.300 approved

### Composition

HDPE/EBS/aluminum oxide

### Renewable Carbon Index

>90%

### Recommended Addition Levels

0.5-1.5% (on total formula weight)

### Systems and Applications

Water based, solvent based and energy curable coatings and inks. Industrial coatings (including plastic and metal); stains, sealers and varnishes; wood coatings; printing inks and OPV's (including flexo and gravure); powder coatings; can, container, and coil coatings; rubber additives.

### Typical Properties\*

|                            | <u>Superslip 6515AL</u> |
|----------------------------|-------------------------|
| Mean Particle Size (µm)    | 3.5 - 5.5               |
| Maximum Particle Size (µm) | 15.56                   |
| Melting Point °C           | 139 - 145               |
| Density @ 25 °C (g/cc)     | 0.99                    |
| NPIRI Grind                | 1.0 - 2.0               |

This product is also available as a water based wax dispersion - Microspersion 6515AL-40

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## AquaPoly 511

A unique micronized oxidized polyethylene wax composite for improved scuff and mar resistance in aqueous and solvent based systems

### Features and Benefits

- A synergistic blend of high molecular weight waxes including polyethylene
- Provides optimum levels of scuff and mar resistance along with excellent slip and mobility
- Easy to disperse fine powder that can be incorporated into aqueous paints, inks and coatings with high speed mixing
- Compare to AquaPolyfluor 411 (without PTFE)

### Composition

Modified oxidized HDPE

### Recommended Addition Levels

0.5-2.0% (on total formula weight)

### Systems and Applications

Solvent and water based coatings. Industrial coatings (including plastic, metal and leather); architectural wall and trim paints; stains, sealers and varnishes; wood coatings; printing inks and OPV's (including flexo and gravure); coil coatings.

### Typical Properties\*

|                            | <u>AquaPoly 511</u> |
|----------------------------|---------------------|
| Mean Particle Size (µm)    | 6.0 - 8.0           |
| Maximum Particle Size (µm) | 22.00               |
| Melting Point °C           | 117 - 123           |
| Density @ 25 °C (g/cc)     | 0.97                |
| NPIRI Grind                | 2.5 - 3.5           |

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## AquaBead® 519

A specially formulated wax composite that provides immediate water repellency and weather resistance

### Features and Benefits

- Imparts immediate water beading
- Provides long-lasting water repellency
- Contains lower molecular weight polymers for faster water beading in water based and solvent based penetrating stains
- Combine with silica to improve weather resistance

### Composition

Hydrophobically modified synthetic wax

### Recommended Addition Levels

1.0-4.0% (on total formula weight)

### Systems and Applications

Water based and solvent based coatings. Industrial coatings (including metal and masonry); stains, sealers and varnishes; wood coatings; automotive polishes.

### Typical Properties\*

|                            | <u>AquaBead 519</u> |
|----------------------------|---------------------|
| Mean Particle Size (µm)    | 5.5 - 8.0           |
| Maximum Particle Size (µm) | 22.00               |
| Melting Point °C           | 126 - 132           |
| Density @ 25 °C (g/cc)     | 0.94                |
| NPIRI Grind                | 2.0 - 3.0           |

This product is also available as a water based wax dispersion - Microspersion 519

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## MicroKlear 418

Micronized T-1 prime yellow refined carnauba wax for surface slip and scratch resistance with excellent clarity and gloss retention. Also useful as an additive in seed coatings

### Features and Benefits

- Provides surface lubricity and scratch resistance
- Does not adversely affect clarity or gloss
- 100% natural and biodegradable
- Easy to disperse fine powder that can be incorporated with high speed mixing
- In seed coatings, improves anti-blocking, flow, and abrasion resistance

### Composition

Carnauba wax

### Renewable Carbon Index

100%

### Recommended Addition Levels

1.0-3.0% (on total formula weight)

### Systems and Applications

Water based, solvent based and energy curable coatings and inks. Industrial coatings (including plastic, metal and leather); wood coatings; printing inks and OPV's (including flexo and gravure); interior and exterior can and container coatings; coil coatings; seed coatings.

### Typical Properties\*

|                            | <u>MicroKlear 418</u> |
|----------------------------|-----------------------|
| Mean Particle Size (µm)    | 6.0 - 8.0             |
| Maximum Particle Size (µm) | 22.00                 |
| Melting Point °C           | 81 - 86               |
| Density @ 25 °C (g/cc)     | 1.00                  |
| NPIRI Grind                | 2.0 - 3.5             |
| Color                      | Light Yellow          |

This product is also available as a water based wax dispersion - Microspersion 418-40

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